WONG JING TAO JOANNE

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SUMMARY

Applied AI developer in computer vision and multimodal sensing, with a seven-month ML research internship at Deakin University's IISRI (RGB-IR fire segmentation, EEG/PPG-driven VR stress monitoring). Final-year Big Data Analytics student at Universiti Teknologi PETRONAS (CGPA 3.65, minor in Accounting & Finance), graduating December 2026. Seeking AI/ML engineer and data scientist roles.

EDUCATION

Bachelor of Information Technology (Honours), Big Data Analytics | Universiti Teknologi PETRONAS

Minor: Accounting & Finance | CGPA 3.65/4.00 | Expected graduation: December 2026

ICAEW Certificate in Finance, Accounting and Business (CFAB), in progress

EXPERIENCE

Machine Learning Intern | Deakin University, IISRI, Australia | Sep 2025 – Apr 2026

Multimodal RGB-IR Wildfire Segmentation

- Designed and implemented HybridModel, an asymmetric dual-stream RGB-IR segmentation network (YOLOv12n + EfficientNet-B2) with spatial gate fusion (per-pixel, per-channel RGB/IR weighting), edge-aware encoders and CBAM, trained with Focal (hard-pixel mining), Dice and Tversky losses for extreme class imbalance.
- Evaluated on two wildfire benchmarks (Corsican Fire Database, 100 pairs; RGB-T wildfire dataset, 1,367 pairs), reaching 91.95% and 88.49% F1 on their validation sets.

Real-Time Stress Monitoring in VR

- Built a real-time stress monitoring pipeline on the Galea VR headset, computing EEG beta/alpha power ratio and PPG-derived heart rate, with live visualization.
- Classified stress state using literature-based thresholds on beta/alpha ratio (below 1.0 relaxed, above 1.3 high stress) and heart rate (60-80 BPM relaxed, above 100 BPM high stress).
- Designed a bio-responsive VR interface that adapts to the detected stress state in real time.

PROJECTS

Hand-Action Recognition AR for ROV Safety Training | Final Year Project | May 2026 – Dec 2026 (ongoing)

- Built a MediaPipe Hands pipeline that classifies 4 ROV training actions (screw, unscrew, pull battery cap, no action), and compared 9 configurations across Random Forest, LSTM and GRU.
- Feature engineering (wrist-relative landmarks, velocity, BlueROV2 detections) raised accuracy from 50% to 75% on 8 held-out clips; chose Random Forest over LSTM and GRU, which did not beat it on this small dataset.
- Integrated HoloLens 2 eye-gaze (MRTK3, OpenXR) with real-time UDP streaming to deliver recognition feedback during training.

Commercial Team ROI Dashboard | KABEL DXP, YEI 3.0 Youth Innovation Sandbox | Sep – Oct 2026

- Built a role-based performance dashboard prototype across 4 levels (Director to Employee) on synthetic HR and commercial data, with ROI on the full cost base and explainable KPI scores.
- Added LLM-generated KPI recommendations (Claude API, rules-based fallback) behind a manager-approval step; Supabase schema with row-level security, unit tests, deployed on Vercel.

ECCO: Interprofessional VR Training Platform | INNOCOM AID National Innovation Competition

- Co-designed, in a team of 6, a multi-user VR platform concept for joint emergency-response training between medical and paramedic teams, including the system architecture and educational framework.
- Pitched the concept's technical feasibility and training applications in the online national competition and won the Best Presenter Silver Award.

SKILLS

Languages and ML: Python, C#, SQL (Postgres), PyTorch, OpenCV, MediaPipe, Ultralytics YOLOv12, Albumentations, LLM API integration (Claude API)

Data: Pandas, NumPy, scikit-learn, KPI and ROI analysis

Tools: Git, Supabase, Vercel

LEADERSHIP

Project Director | Festival of Colours of the World (FESCO) 2025, UTP

- Directed an international multicultural festival for 2,400+ attendees and 500+ participants across 32 competing teams, leading a 130+ member committee and managing RM20,000 in sponsorship.

President | UTP Performing Arts Group (UPAG)

- Led a 50+ member student organization, overseeing strategy, events and university performances.

AWARDS

- IEEE Academia-Industry Summer School (AISS) 2025 Scholarship Recipient, Swinburne University of Technology.
- Best Presenter Silver Award, ECCO, INNOCOM AID Innovation Competition
- Dean's List, every semester, Universiti Teknologi PETRONAS
- Top 3 Finalist, Distinguished President Award, Student Excellence Awards 2025
- Cheonan World Dance Festival, South Korea: Gold (2023); Bronze and Best Choreography (2024)